

ECE 215: Circuit Theory and Devices (CTD)

Mid-Sem Exam (Set A)

Date: September 27, 2025	Duration: 1 hour	Total points: 30
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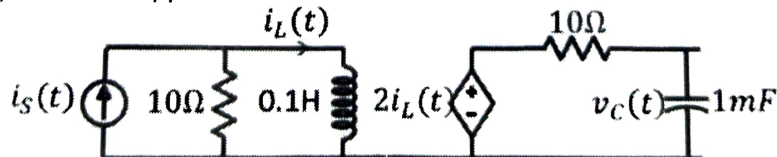
Instructions

- Write your set in the answer sheet.
- Remember to write units in all your answers.
- Clearly mention the question number you are attempting. Otherwise, zero marks will be given.

Questions 1 [CO1] [10 points]

Consider the circuit shown below, where the source is $i_S(t) = 10\cos(50t)$ A. Assume that none of the elements carry any energy at the initial conditions.

- Find Thévenin equivalent across the terminals of the capacitor (i.e., exclude the capacitor while calculating Thévenin equivalent).
- Find the time domain $i_L(t)$ and $v_C(t)$. Make use of Thévenin equivalent model for these calculations, wherever applicable.



Questions 2 [CO1, CO2] [10 points]

A balanced three-phase system having a line voltage of 240 V rms contains a delta-connected load of $(12 + j)$ k Ω per phase and also a Y-connected load of $(5 + j3)$ k Ω per phase.

- Find the line current magnitude.
- Find the average power taken by the combined load.
- Calculate the power factor of the load.

Questions 3 [CO1, CO2] [10 points]

For the circuit shown below, assume the source operates at a frequency of 100 rad/s.

- Determine the PF at which the source is operating.
- Calculate the apparent power absorbed by each of the three passive elements.
- Compute the average power supplied by the source and the average power delivered to each passive element.

